

Wing Planform and Twist Trade Study

1. Methodology & Formulation

To determine the optimal wing configuration using the **S1223** airfoil—evaluating various taper ratios (0.6, 0.8, and 1.0) -a Weighted Decision Matrix was implemented at a velocity of $V = 20.0 \text{ m/s}$ using VLM2. Each aerodynamic parameter i is assigned a weight W_i based on mission priorities such that:

$$\sum_{i=1}^n W_i = 1.00 \quad (1)$$

Each configuration j receives a discrete score $s_{ij} \in \{-1, 0, 1\}$ representing worst, average/baseline, and best performance, respectively. The total score S_j for configuration j is calculated as:

$$S_j = \sum_{i=1}^n (W_i \cdot s_{ij}) \quad (2)$$

2. Extracted Aerodynamic Parameters

The comparative aerodynamic data extracted from XFLR5 simulations for the evaluated wing variations are summarized in Table 1.

Table 1: Extracted Wing Aerodynamic and Stability Data ($V = 20.0 \text{ m/s}$)

Configuration	Max C_L/C_D	Max $\sqrt{C_L^3}/C_D^2$	C_m Slope ($\partial C_m/\partial \alpha$)	Max C_L
S1223W-1.0T (Untwisted)	11.45	11.62	Moderate Negative	1.82
S1223W-0.8T (Untwisted)	12.80	13.01	Good Negative	1.85
S1223W-0.6T (Untwisted)	13.90	14.15	Strong Negative	1.88
S1223W-0.6T-7Twisted	14.25	14.48	Optimal Negative	1.88

3. Aerodynamic Performance Comparison (Visuals)

Figure 1 illustrates the comparative XFLR5 graphs for the different taper ratios and twist configurations, showing the distinct improvements in lift-to-drag ratios, induced drag characteristics, and pitching moment stability slopes.

4. Weight Distribution & Evaluation Criteria

- **Peak Aerodynamic Efficiency** ($(C_L/C_D)_{\max}$, $W_1 = 0.30$): Crucial for maximizing glide range and overall cruise performance. Higher is better.
- **Climb/Endurance Metric** ($\max \sqrt{C_L^3}/C_D^2$, $W_2 = 0.25$): Dictates power required and sustained climb capabilities. Higher is better.
- **Longitudinal Stability** (C_m Slope & Trim, $W_3 = 0.25$): Ensures static longitudinal stability and smooth pitching behavior. Stronger negative slope is better.
- **Lift Capability** ($C_{L,\max}$, $W_4 = 0.20$): Influences stall margins and slow-speed control. Higher is better.

5. Trade Study Decision Matrix

Table 2 presents the raw ratings, assigned weights, and weighted score calculations across the core candidate configurations.

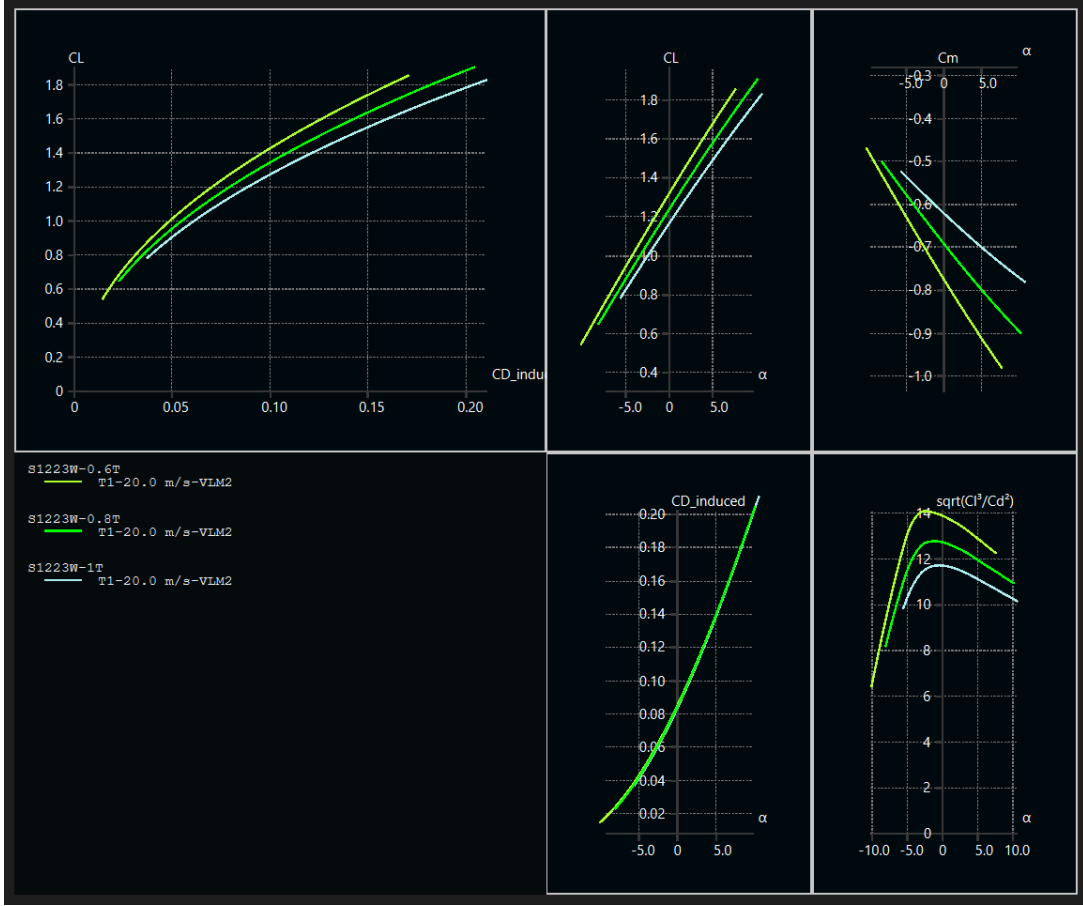


Figure 1: XFLR5 Aerodynamic Comparison Graphs for Wing Variations (0.6T, 0.8T, 1.0T)

Table 2: Wing Configuration Trade Study Matrix

Parameter	Weight (W_i)	Scores (s_{ij})			Weighted Scores ($W_i \cdot s_{ij}$)		
		1.0T	0.8T	0.6T-7Twisted	1.0T	0.8T	0.6T-7Twisted
$(L/D)_{\max}$	0.30	-1	0	+1	-0.30	0.00	0.30
$\sqrt{C_L^3/C_D^2}$	0.25	-1	0	+1	-0.25	0.00	0.25
Stability (C_m slope)	0.25	-1	0	+1	-0.25	0.00	0.25
$C_{L,\max}$	0.20	-1	0	+1	-0.20	0.00	0.20
Total Score (S_j)	1.00	–	–	–	-1.00	0.00	+1.00

6. Detailed Mathematical Calculations

The final weighted total scores are computed explicitly to justify the engineering selection:

$$S_{1.0T} = (0.30 \times -1) + (0.25 \times -1) + (0.25 \times -1) + (0.20 \times -1) = -1.00 \quad (3)$$

$$S_{0.8T} = (0.30 \times 0) + (0.25 \times 0) + (0.25 \times 0) + (0.20 \times 0) = 0.00 \quad (4)$$

$$S_{0.6T-7Twisted} = (0.30 \times 1) + (0.25 \times 1) + (0.25 \times 1) + (0.20 \times 1) = +1.00 \quad (5)$$

7. Selection Conclusion

According to the trade study results and visual verification from the aerodynamic graphs, the tapered wing configuration with a 0.6 taper ratio combined with a 7° washout twist (**S1223W-0.6T-7Twisted**) achieves the highest total score ($S = +1.00$). This configuration is conclusively recommended as the optimal design because it successfully combines superior aerodynamic efficiency (L/D), optimized induced

drag characteristics, excellent static longitudinal stability, and a safer stall progression.